

Project-Cycle-3

One-page Summary

Group 25

Research question

Is the proportion of current cigarette use different between male and female students?

Group definition

- Group variable: WhatIsYourSex
- Definition: This variable classifies students into two groups for analysis, where 0 = Female and 1 = Male.

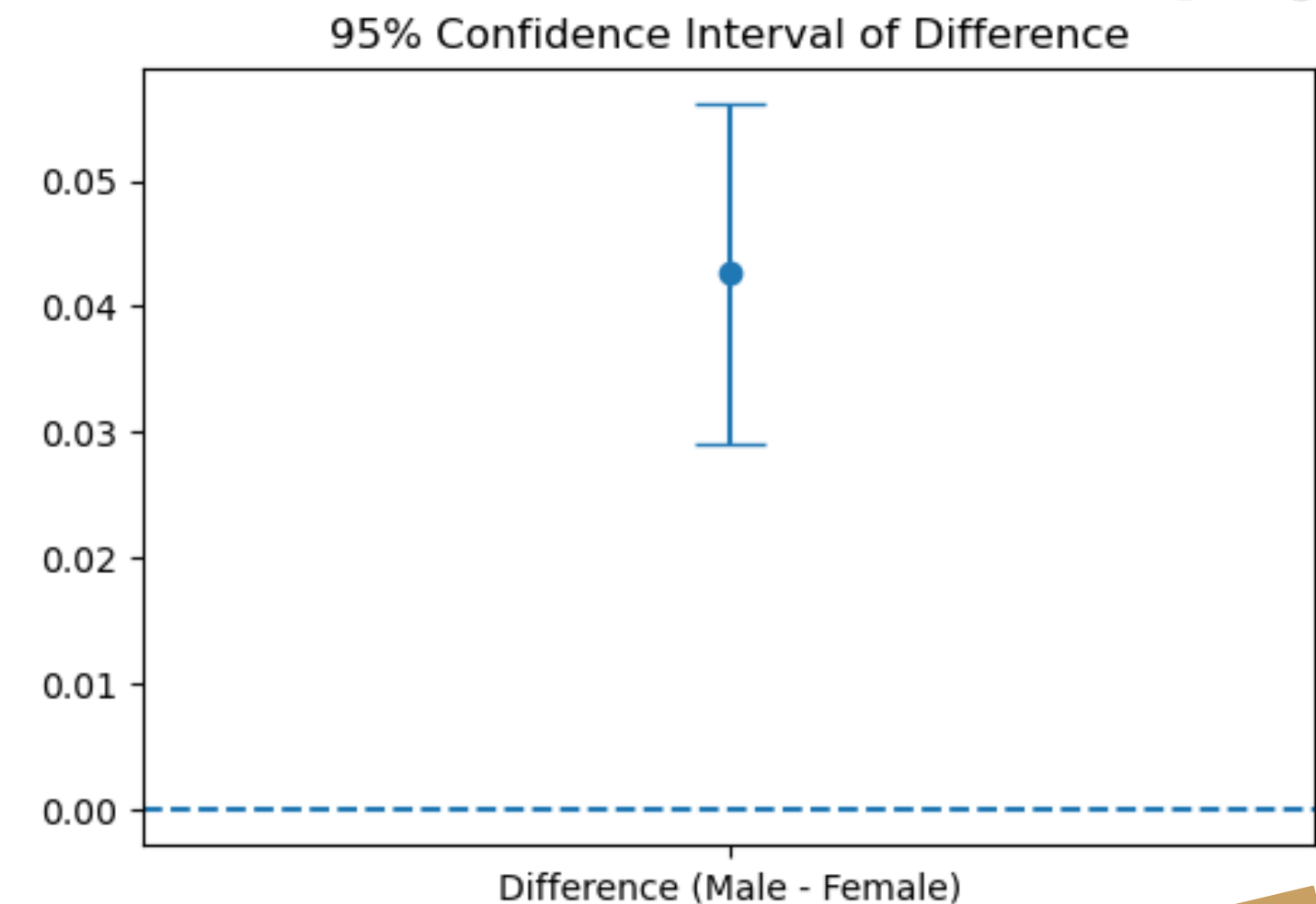
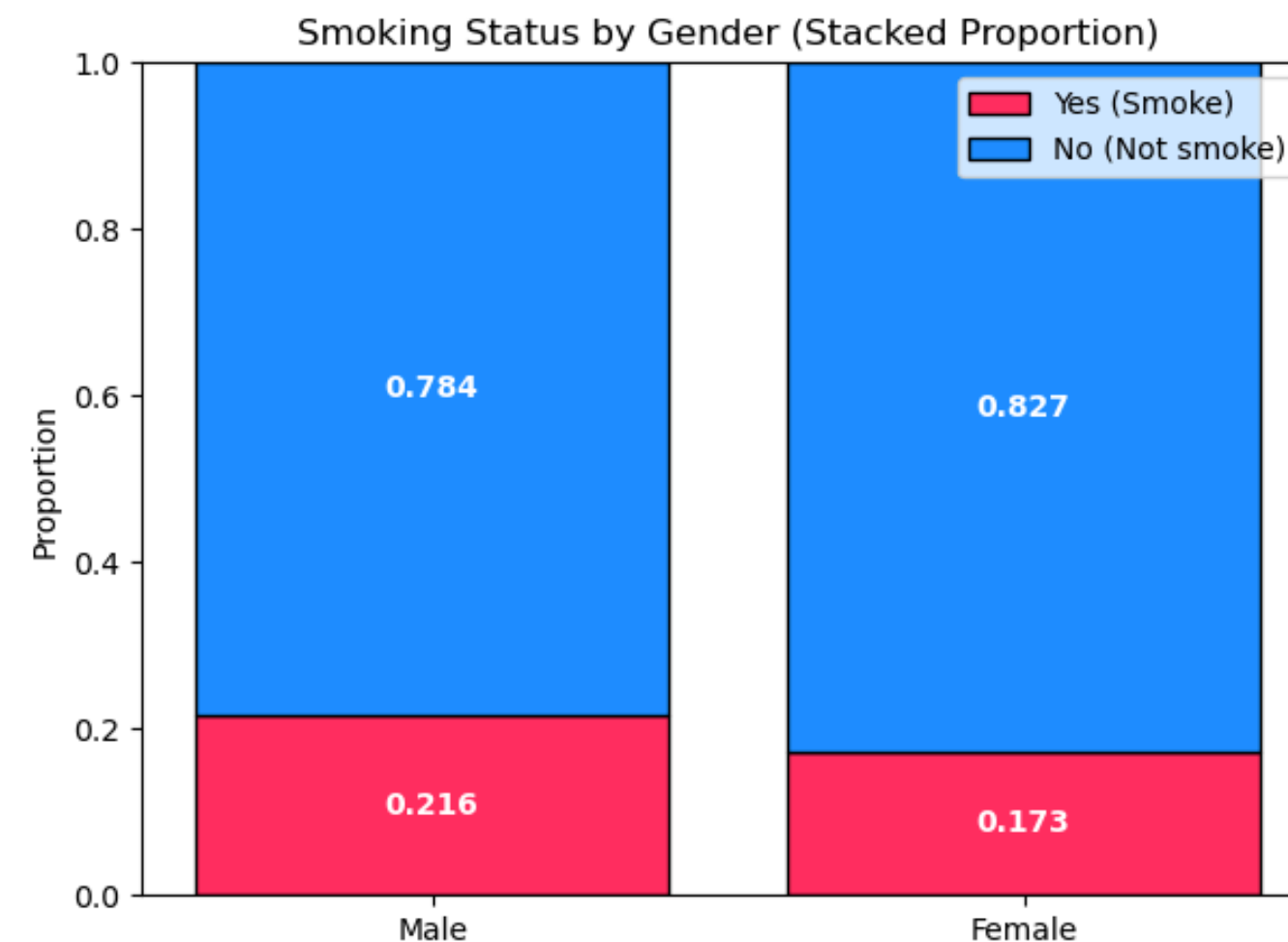
Response variable

- Response variable: CurrentCigaretteUse
- Definition: This variable indicates whether students currently smoke cigarettes within the past 30 days, where 0 = non-smoker and 1 = current smoker.

Method

Two-proportion z-test

Figures



Key result

- Difference in proportions ($\hat{p}_{\text{male}} - \hat{p}_{\text{female}} = 0.043$) was positive.
- 95% CI did not include 0, suggesting a meaningful difference between the two groups.
- The two-proportion z-test produced a p-value ($5.14e-10$) < 0.05 , leading to rejection of the null hypothesis ($H_0: p_{\text{male}} = p_{\text{female}}$).
- Male students had a higher smoking proportion than female students.

Conclusion

There is sufficient statistical evidence at the 0.05 significance level to conclude that the proportion of current cigarette use differs between male and female students.

